Masked Pretraining Talk — Speaker Script

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Read one section per slide. At a normal speaking pace the whole script runs about 4 minutes, inside the 3–5 minute limit.

Slide 1 · Title

Hi, I'm Samantha Salas, and together with John Edwards we're studying how much masked-image pretraining helps object detection when labeled data is limited.

I'll cover our question, how we designed the experiment, where the project stands, and what we expect to find.

Slide 2 · Motivation

Object detectors normally learn only from human-drawn boxes, and drawing those boxes is the expensive part. With few labels, accuracy drops sharply.

Masked image modeling offers another signal: hide most of an image, have the backbone reconstruct it, and learn from unlabeled images before ever seeing a box.

Our question is how much that pretraining buys over random initialization, and how the benefit changes as labels shrink. We expect the gap to widen at small label fractions, and to be larger for a Vision Transformer, since it lacks the convolutional inductive bias and leans more on what pretraining provides.

Slide 3 · A controlled comparison

Prior papers report these gains under different schedules and detection heads, and they pretrain on ImageNet, so the numbers aren't directly comparable.

We control all of that. Both detectors share one Detectron2 protocol. Pretraining uses the same COCO images the detector later trains on, so no outside data enters. And we sweep the labeled fraction from ten to one hundred percent.

Slide 4 · Experimental design

At every label fraction we train the detector twice. In the pretrained arm, the backbone first learns from unlabeled images with MAE or SparK. In the random arm, it starts from random weights. Everything after that is identical, so the difference in AP is what pretraining bought.

We test two families: a ViT-Tiny with MAE in ViTDet, and a ResNet-18 with SparK in Faster R-CNN. We use tiny backbones to fit our compute. That gives two pretraining runs and forty detection runs.

Slide 5 · Where we are

Here's where things stand. The COCO subset is prepared on UTSA's Arc cluster, eighteen thousand five hundred training images. Detectron2 is built with checkpoint resuming so long jobs survive time limits, and we wrote the tiny ViT and ResNet-18 configs ourselves, since neither ships with the original code.

John's MAE and SparK pretraining runs have finished, and all twenty random-initialization runs are training now. Next, we fine-tune the other twenty detectors from the pretrained weights and evaluate everything on val2017.

Slide 6 · What we will measure

Our main result will be this plot: the AP gain from pretraining at each label fraction. What's shown is our hypothesis, not data.

If the curve falls from left to right, pretraining matters most when labels are scarce. A flat curve means it's a constant boost. And if the ViT curve sits above the ResNet curve, transformers rely more on pretraining. We'll also break AP down by IoU threshold and object size to see where the gain comes from.

Slide 7 · Challenges

A few things could bias the comparison. The biggest is that training from scratch converges slower, so a short schedule would exaggerate pretraining's benefit; we'll report AP over the course of training to make that visible.

At ten percent of labels, sampling noise can be as large as the effect, so we repeat those runs with different seeds. Porting two codebases into Detectron2 risks silently changing one arm, so we check settings against the originals. And because of compute, our absolute AP will be low, so we focus on the gap, not the raw number.

Slide 8 · Summary

To sum up: with one shared protocol, two backbone families, and ten label fractions, we'll get a direct measure of what unlabeled images are worth when labeled boxes are scarce. Thanks for watching.